

Do Politicians' Words Have a Dollar Value?

Measuring the Impact of Climate Policy Speeches on Green Bond Prices in
India, Indonesia, and Thailand

Nahian Ibnat

MA in Economics, Data & Policy (2 years)

Supervisor:

Zoltan Csaba Toth

March 30, 2026

Contents

Abstract	3
1 Introduction: The Greenium in Emerging Economies	4
2 Literature Review	6
2.1 What Exactly Is a Green Bond?	6
2.2 Green Bonds as Sovereign Statecraft: The IPE Perspective	7
2.3 Strategic Importance in the Transition Giants	7
2.4 The Debate: Market Efficiency vs. Political Credibility	7
2.5 Sentiment: The Role of ClimateBERT	8
2.6 Methodological Contribution and Scope	8
2.7 Research Hypotheses	8
3 Data	9
3.1 Data Sources and Technical Justification	9
3.2 Financial Yield Data: AsianBondsOnline (ADB)	9
3.3 Political Event Tracking: The GDELT Project	9
3.4 Linguistic Sentiment Analysis: ClimateBERT	10
4 Methodology	10
4.1 Research Design: The Twin-Bond Quasi-Experiment	10
4.2 The NLP Pipeline: Quantifying Political Signaling	11
4.3 Econometric Strategy: Staggered Difference-in-Differences	11
4.4 Event Study: Leads, Lags, and Pre-announcement Drift	12
4.5 Identification of Political Shocks and Case Selection	12
5 Data Analysis and Empirical Results	13
5.1 Descriptive Statistics	13
5.2 Sentiment Analysis Results	14
5.3 Regression Outputs	14
5.4 Parallel Trends Test	14
6 Discussion and IPE Interpretation	14
6.1 Bridging the Credibility Gap	14
6.2 The Cost of “Cheap Talk”	14
6.3 Global vs. Local Pressure	15
7 Conclusion and Policy Recommendations	15
7.1 Summary of Findings	15
7.2 Policy Recommendations	15

7.3 Limitations and Future Research	15
References	16

Abstract

Sovereign green bonds have emerged as critical instruments for financing sustainable development in emerging markets. However, the *Greenium*—defined as the yield discount investors accept for environmentally aligned assets—remains highly volatile, particularly in Asian economies. This study investigates whether political climate sentiment constitutes a causal driver of secondary-market Greenium dynamics in India, Indonesia, and Thailand over the period 2020–2025. To quantify political signaling, this paper employs a Natural Language Processing (NLP) framework using ClimateBERT to extract high-frequency sentiment from domestic news headlines obtained via the GDELT Project. Let S_t denote the daily climate sentiment score and G_{it} the Greenium (yield spread) for bond i at time t . The empirical strategy estimates the causal relationship:

$$G_{it} = \alpha + \beta S_t + \gamma X_{it} + \varepsilon_{it}$$

within a staggered Difference-in-Differences (DiD) and event study framework, enabling identification of exogenous *political sentiment shocks*. The results provide evidence that positive climate sentiment is associated with a widening Greenium, reflecting a *Credibility Premium* in sovereign borrowing costs. In contrast, conflicting political signals—captured as *signaling noise* across government institutions—generate a measurable *Credibility Gap*, increasing yields and imposing a trust-related penalty on public debt. These findings contribute to the literature by demonstrating that political communication is not merely rhetorical, but a priced financial variable. In emerging markets, where policy credibility is often contested, coherent and consistent climate signaling functions as a strategic fiscal tool that can materially reduce the cost of capital during the green transition.

Keywords: Green bonds, Greenium, political sentiment, ClimateBERT, Difference-in-Differences, sovereign debt, emerging markets, climate finance

1 Introduction: The Greenium in Emerging Economies

Imagine a world in which a single speech by a Prime Minister does not merely influence voters, but directly lowers the national debt. In the high-stakes world of international finance, this is increasingly becoming a reality through an instrument known as the Green Bond. While traditional bonds are simply loans that governments use to finance general expenditures, a Green Bond is a loan with a commitment. The issuing government pledges that all proceeds will be allocated exclusively to environmentally sustainable projects, such as renewable energy infrastructure or environmental remediation.

In return for this transparency, environmentally conscious investors are often willing to accept a lower yield. This yield discount is referred to as the *Greenium* (Zerbib, 2019). For large sovereign issuers, even a marginal reduction in borrowing costs—measured in basis points—can translate into substantial fiscal savings, which can then be reinvested into the broader economy.

Understanding the dynamics of these instruments is particularly important for the Transition Giants of Asia: India, Indonesia, and Thailand. These countries provide an ideal empirical setting, as they embody a structural paradox. They are large, rapidly growing economies that remain heavily dependent on coal, yet they have also emerged as leading issuers of sovereign green bonds. In India, the inaugural 2023 green bond issuance reduced borrowing costs by approximately 5–6 basis points. Indonesia pioneered the world’s first sovereign Green Sukuk, while Thailand has developed one of the most advanced green yield curves in the region (Panizza et al., 2025).

For these economies, green bonds provide fiscal space by enabling the financing of costly energy transitions without significantly increasing debt burdens (Kling et al., 2021). When governments credibly signal commitment to climate objectives, investors reward them with lower financing costs. However, when that credibility is undermined, the Greenium diminishes or disappears entirely, increasing the cost of capital.

Trust, however, is difficult to quantify and easy to erode. In many emerging markets, there exists a persistent disconnect between what can be termed *Green Talk* and *Brown Action*. This study addresses a central puzzle: does the financial market respond to high-frequency political communication, and if so, how does political credibility translate into measurable borrowing costs?

To conceptualize this mechanism, I introduce two key constructs. The *Credibility Premium* refers to the reward investors grant when political rhetoric is consistent with policy action; under such conditions, perceived risk declines and the Greenium widens (Lau et al., 2024). Conversely, the *Credibility Gap* captures the penalty imposed when governments send conflicting signals. For example, if a Ministry of Finance issues a green bond while the Ministry of Energy simultaneously approves new coal infrastructure, investors interpret this as *signaling noise* (Neumann, 2025). The resulting uncertainty increases perceived risk, leading to a contraction of the Greenium and higher borrowing costs.

Existing empirical literature on green bonds largely relies on low-frequency data, such as monthly indices or aggregate macroeconomic indicators. While informative for long-term trends, these approaches fail to capture the high-frequency nature of political communication. In reality, political signaling operates at the pace of daily news cycles. Consequently, there remains limited understanding of how discrete political events or daily sentiment shifts influence sovereign borrowing costs in real time.

This paper addresses this gap by integrating Data Science methods into the study of International Political Economy (IPE). Specifically, I employ ClimateBERT, a Natural Language Processing (NLP) model trained to interpret climate-related discourse (Webersinke et al., 2022). Using data from the GDELT Project (Leetaru & Schrod, 2013), the model extracts daily political sentiment from news coverage and official communication. By aligning this high-frequency sentiment measure with daily bond market data, the analysis identifies the causal impact of political communication on sovereign yield spreads.

The primary research question is therefore:

How does political climate sentiment causally influence the sovereign Greenium in India, Indonesia, and Thailand, and what is the measurable financial cost of a Credibility Gap?

This study also addresses two sub-questions:

- Can NLP models such as ClimateBERT capture market-relevant information that traditional low-frequency policy indices overlook?
- Does the market reward political consistency with a Credibility Premium, and if so, what is its magnitude in basis points?

The central thesis of this paper is that political credibility constitutes a priced financial asset. In emerging markets, political signaling is not merely informational but plays a direct role in determining the cost of sovereign borrowing. Consistent and credible climate communication functions as a strategic fiscal tool, enabling governments to secure lower financing costs through a Credibility Premium.

To empirically test this argument, the study employs a Staggered Difference-in-Differences (DiD) framework. Conceptually, this approach compares a sovereign green bond (treatment group) with a conventional maturity-matched “twin” bond (control group) issued by the same government. By focusing on major exogenous policy events—such as India’s Sovereign Green Bond Framework (2022), Indonesia’s Just Energy Transition Partnership (JETP), and Thailand’s Green Taxonomy updates—the analysis isolates the causal effect of political sentiment on yield differentials. This design effectively controls for global macroeconomic shocks, such as fluctuations in oil prices or international interest rates, allowing the analysis to focus on the political signal itself.

The contribution of this research lies in its integration of technological, economic, and political perspectives. By demonstrating that Artificial Intelligence can be used to measure sovereign credibility in real time (Dragotto et al., 2025), the study extends the literature beyond traditional, low-frequency, and predominantly Western-focused analyses. It provides the first high-frequency empirical evidence on Greenium dynamics in Asian emerging markets.

More broadly, the findings suggest that political communication has direct financial consequences. Harmonized and credible climate signaling leads to a widening Greenium, confirming the presence of a Credibility Premium. In contrast, inconsistent policy messaging generates a measurable trust penalty, reducing the borrowing advantage associated with green bonds. These results imply that environmental credibility is not only a normative concern but a central component of sovereign debt management.

The remainder of the paper is structured as follows. Chapter 2 reviews the existing literature on green bonds and political credibility. Chapter 3 describes the data and the construction of the sentiment measures using GDELT and ClimateBERT. Chapter 4 outlines the econometric methodology, including the Twin-Bond design and the DiD framework. Chapter 5 presents the empirical results, identifying key political shocks and their market effects. Chapter 6 adds the discussion and interpretation of the results while explaining the importance of the results. Finally, Chapter 7 concludes with policy recommendations for mitigating the Credibility Gap and enhancing sovereign credibility in green finance.

2 Literature Review

2.1 What Exactly Is a Green Bond?

To evaluate the financial impact of political rhetoric, it is first necessary to define the instrument at the center of the analysis. A green bond is a fixed-income instrument in which the issuer contractually commits to allocating net proceeds toward Eligible Green Projects, such as renewable energy or sustainable water management (ICMA, 2022). While structurally identical to a standard bond, it is distinguished by this procedural conditionality, typically enforced through a Use of Proceeds clause (ICMA, 2022).

The primary financial metric used to evaluate these bonds is the *Greenium* (Green Premium), defined as the yield difference between a green bond and a maturity-matched conventional “twin” bond issued by the same sovereign. Formally, this can be expressed as:

$$\text{Greenium}_{it} = y_{it}^{\text{green}} - y_{it}^{\text{conv}}$$

When this value is negative, the green bond trades at a lower yield (and thus a higher price) than its non-green counterpart. This spread is not merely a mathematical construct; it represents a market-based valuation of environmental credibility. Zerbib (2019) pioneered the matching method to isolate this spread, arguing that the Greenium is driven by investors’

pro-environmental preferences, whereby market participants accept lower financial returns in exchange for non-pecuniary environmental benefits.

2.2 Green Bonds as Sovereign Statecraft: The IPE Perspective

From the perspective of International Political Economy (IPE), sovereign green bonds are not neutral financial instruments but strategic tools through which states engage with global capital markets. Gabor (2021) conceptualizes this dynamic as the *Wall Street Consensus*, whereby emerging markets are required to adopt internationally recognized environmental and financial standards in order to maintain access to global capital.

For Transition Giants such as India, Indonesia, and Thailand, issuing a green bond constitutes an act of sovereign signaling. It communicates to the global financial core that the state is aligned with contemporary environmental governance norms. However, this signaling function introduces tension between international expectations and domestic political realities. Kling et al. (2021) show that while green bonds provide fiscal relief by lowering climate-related risk premia, they also impose constraints by requiring alignment with global disclosure and regulatory standards, which may conflict with domestic development priorities.

2.3 Strategic Importance in the Transition Giants

The importance of the Greenium in Asia is further complicated by what IPE scholars describe as the *Minerals-Energy Complex* (Rethel & Sinclair, 2014). In economies such as India and Indonesia, political stability often depends on coal-dependent industries, labor groups, and state-owned enterprises that resist rapid decarbonization.

This creates a structural *Double Bind*: the Ministry of Finance must signal Green Ambition to international investors, while the Ministry of Energy must maintain Brown Stability to satisfy domestic stakeholders (Neumann, 2025). This internal contradiction generates what this study conceptualizes as *Signaling Noise*—a key source of uncertainty in sovereign bond pricing.

Empirically, India's 2023 inaugural green bond issuance demonstrated measurable borrowing cost reductions (Panizza et al., 2025). Indonesia's sovereign Green Sukuk expanded Islamic finance while signaling commitment to the Just Energy Transition Partnership (JETP), and Thailand has leveraged green bonds to support its Bio-Circular-Green (BCG) strategy. Across all cases, the Greenium reflects a reduction in climate-risk premia, making the energy transition financially feasible (Kling et al., 2021).

2.4 The Debate: Market Efficiency vs. Political Credibility

There is an ongoing academic debate regarding the persistence of the Greenium. The *Preference School* argues that regulatory pressures and ESG mandates will sustain excess demand for green assets, ensuring a persistent yield discount (Baker et al., 2022). In contrast, the *Efficiency*

School, led by Larcker and Watts (2020), contends that no Greenium should exist, as green and conventional bonds share identical default risk.

This study contributes to this debate by introducing *policy credibility* as a decisive factor (Lau et al., 2024). Within the IPE framework, credibility functions as a critical intangible asset. When a government is perceived as having sufficient institutional capacity to implement its climate commitments, the market rewards it with a *Credibility Premium*. Conversely, weak or inconsistent policy signals result in a *Trust Penalty*.

2.5 Sentiment: The Role of ClimateBERT

This research links financial theory with Data Science by employing ClimateBERT to measure political credibility in real time. ClimateBERT, developed by Webersinke et al. (2022), is a specialized Natural Language Processing (NLP) model designed to detect climate-related meaning beyond simple keyword matching.

Unlike traditional models, ClimateBERT captures contextual nuance, distinguishing between vague statements and credible commitments. By applying this model to high-frequency news data from the GDELT Project, the study moves beyond low-frequency indices such as the Economic Policy Uncertainty index. This enables the quantification of *Signaling Noise*, defined as the divergence between Green Talk and Brown Action (Neumann, 2025). This approach aligns with Dragotto et al. (2025), who find that daily climate awareness significantly influences financial market dynamics.

2.6 Methodological Contribution and Scope

This study contributes to the literature by introducing the concept of the *Credibility Gap*. Existing research typically assumes that once a bond is labeled green, its classification remains stable. This paper challenges that assumption by testing whether inconsistencies between political commitments and policy actions affect bond pricing.

I hypothesize that such inconsistencies generate a Trust Penalty, leading to a contraction of the Greenium. Conversely, alignment between rhetoric and policy results in a Credibility Premium. Using a Staggered Difference-in-Differences (DiD) framework with leads and lags, the analysis captures *Pre-announcement Drift*, where markets anticipate policy signals prior to formal announcements (MacKinlay, 1997).

By focusing on Asian emerging markets, this research provides the first causal evidence on how localized political communication affects sovereign borrowing costs.

2.7 Research Hypotheses

Based on the theoretical tensions identified in the literature—particularly the conflict between the Wall Street Consensus and domestic coal dependence—this study tests three primary hypotheses.

Hypothesis 1: The Credibility Premium (Sentiment Effect)

An increase in positive climate sentiment, as measured by ClimateBERT, is negatively correlated with the green bond yield spread, resulting in a widening Greenium.

Hypothesis 2: The Signaling Noise Penalty (Credibility Gap)

The presence of contradictory policy signals reduces the yield-lowering effect of positive sentiment, generating a Trust Penalty and shrinking the Greenium.

Hypothesis 3: The Information Discovery Effect (Pre-announcement Drift)

In a high-frequency information environment, changes in the Greenium occur prior to formal announcements (e.g., at $t - 2$), reflecting market anticipation of political signals captured through news sentiment.

3 Data

3.1 Data Sources and Technical Justification

The empirical strength of this study relies on the triangulation of three authoritative data sources. Each source is selected to ensure high temporal resolution and geographic specificity for the Transition Giants of Asia.

3.2 Financial Yield Data: AsianBondsOnline (ADB)

Daily Yield-to-Maturity (YTM) data for sovereign green and conventional bonds are obtained from AsianBondsOnline, a platform managed by the Asian Development Bank (ADB). The ADB is widely recognized as a reliable source for harmonized fixed-income data in the Asia-Pacific region, providing verified secondary market records that offer superior coverage of emerging markets compared to private commercial databases (Panizza et al., 2025).

The use of YTM, rather than simple bond prices, allows the analysis to account for coupon structures and time-to-maturity, thereby providing a more accurate measure of the effective cost of sovereign borrowing (Zerbib, 2019).

3.3 Political Event Tracking: The GDELT Project

To capture high-frequency political signaling, this study utilizes the GDELT 2.0 (Global Database of Events, Language, and Tone) (Leetaru & Schrodt, 2013). GDELT is a real-time, open-source platform that monitors global news coverage in over 100 languages.

This choice is particularly important for the study's International Political Economy (IPE) focus. While many conventional datasets emphasize English-language international media, GDELT captures localized, vernacular news sources in Hindi, Indonesian, and Thai. This enables the identification of domestic political dynamics and institutional frictions that are often underrepresented in global datasets (Dragotto et al., 2025).

3.4 Linguistic Sentiment Analysis: ClimateBERT

The linguistic analysis is conducted using ClimateBERT (Webersinke et al., 2022), a transformer-based Natural Language Processing (NLP) model specifically trained on climate-related disclosures. Unlike general-purpose language models, ClimateBERT is designed to distinguish between vague environmental claims and concrete, credible policy commitments. This distinction is crucial for reducing measurement error when quantifying political climate sentiment. As highlighted by Webersinke et al. (2022), domain-specific models significantly improve the accuracy of sentiment classification in environmental contexts.

By combining ClimateBERT with high-frequency GDELT data, the study constructs a daily measure of political climate sentiment, enabling the identification of signaling dynamics that would be obscured in lower-frequency datasets.

4 Methodology

4.1 Research Design: The Twin-Bond Quasi-Experiment

The central challenge in empirical finance is the identification problem—isolating the impact of a specific variable, such as political sentiment, from the multitude of global macroeconomic factors that move bond markets daily. To address this, the study adopts a quasi-experimental Twin-Bond design based on the matching methodology pioneered by Zerbib (2019).

For every sovereign green bond issued by India, Indonesia, and Thailand, a maturity-matched conventional “twin” bond is identified. By selecting bonds from the same national issuer, denominated in the same currency, and with nearly identical maturity profiles, this framework constructs a treatment and control structure. The green bond represents the treated asset, while the conventional bond serves as the control.

This comparison is operationalized through the Greenium, which serves as the dependent variable:

$$\Delta y_{i,t} = YTM_{i,t}^{GB} - YTM_{i,t}^{CB}$$

where $\Delta y_{i,t}$ denotes the yield spread between the green bond (GB) and its conventional twin (CB) for bond pair i at time t .

If $\Delta y_{i,t} < 0$, the green bond trades at a lower yield than its conventional counterpart, implying a borrowing advantage for the sovereign issuer. Because the comparison is constructed within the same issuer, this design effectively eliminates confounding factors such as sovereign default risk, currency fluctuations, and exposure to global financial conditions. As a result, changes in the Greenium can be interpreted as reflecting the market valuation of environmental credibility.

4.2 The NLP Pipeline: Quantifying Political Signaling

Unlike traditional IPE research that relies on static or low-frequency policy indices, this study treats political credibility as a high-frequency financial variable. To quantify political signaling, a multi-stage Natural Language Processing (NLP) pipeline is developed.

The key independent variable is the daily climate sentiment score:

$$Sentiment_{c,t} \in [-1, 1]$$

where c indexes country and t denotes time. The primary data source is the GDELT 2.0 database, which captures global news coverage across more than 100 languages, including localized political discourse in emerging markets.

Textual data are processed using ClimateBERT (Webersinke et al., 2022), a transformer-based model trained on climate-related disclosures. The model assigns a continuous sentiment score ranging from -1 (hostile climate sentiment) to $+1$ (ambitious climate sentiment). This allows the construction of a daily measure of political climate communication, capturing both Green Ambition and potential Signaling Noise when rhetoric and policy diverge.

4.3 Econometric Strategy: Staggered Difference-in-Differences

To establish a causal relationship between political sentiment and the Greenium, the study employs a staggered Difference-in-Differences (DiD) framework combined with a fixed-effects panel specification.

The baseline regression model is:

$$\Delta y_{i,t} = \alpha_i + \delta_t + \beta_1 Post_{c,t} + \beta_2 Sentiment_{c,t} + \gamma X_{c,t} + \varepsilon_{i,t}$$

where:

- $\Delta y_{i,t}$ is the Greenium for bond pair i at time t
- $Post_{c,t}$ is a dummy equal to 1 after a major political shock in country c
- $Sentiment_{c,t}$ is the daily climate sentiment score
- $X_{c,t}$ represents additional control variables
- α_i denotes bond-pair fixed effects
- δ_t denotes time fixed effects
- $\varepsilon_{i,t}$ is the error term

Interpretation of coefficients:

- β_1 captures the average structural shift in the Greenium following a major political event. A negative coefficient indicates that such events increase investor confidence and widen the Greenium, thereby lowering the relative borrowing cost of green bonds. Conversely, a positive coefficient suggests that the market interprets the event as inconsistent or non-credible, resulting in a Credibility Gap.
- β_2 measures the marginal effect of daily political sentiment on the Greenium. A negative value implies that more positive climate communication reduces green bond yields relative to conventional bonds, reflecting the presence of a Credibility Premium. A positive coefficient indicates that negative or contradictory sentiment increases borrowing costs through a Trust Penalty.
- α_i controls for all time-invariant characteristics of each bond pair, including structural features of the bonds and issuer-specific risk factors.
- δ_t absorbs common shocks affecting all bonds at time t , such as global interest rate changes or commodity price volatility, ensuring that identification is driven by local political signals.

This specification ensures that causal identification relies on within-pair variation over time, isolating the financial impact of political communication.

4.4 Event Study: Leads, Lags, and Pre-announcement Drift

To analyze the temporal dynamics of political signaling, the study employs an event study framework with leads and lags:

$$\Delta y_{i,t} = \sum_{k=-5}^{+5} \beta_k D_{c,t}^{(k)} + \alpha_i + \delta_t + \varepsilon_{i,t}$$

where $D_{c,t}^{(k)}$ is an indicator for event time relative to a political shock.

The lead coefficients ($k < 0$) test for pre-event trends and detect Pre-announcement Drift, where investors anticipate political developments before official announcements. If these coefficients are statistically insignificant, it supports the validity of the causal design.

The lag coefficients ($k > 0$) capture the persistence of the effect, indicating whether the impact of political signaling results in a temporary market reaction or a sustained shift in borrowing costs. A persistent negative effect reflects a durable Credibility Premium, while a short-lived or reversed effect suggests limited market confidence.

4.5 Identification of Political Shocks and Case Selection

The empirical analysis focuses on major political and regulatory events that are plausibly exogenous to daily bond market movements.

In India, a primary political shock is the January 2023 inaugural sovereign green bond auction. Although the government had released its green bond framework several months earlier, the auction represented the first true price discovery event, where international investors evaluated whether India’s climate commitments justified a lower borrowing cost. By applying the event study framework, the analysis examines whether a Credibility Premium was realized, or whether investor skepticism—driven by continued domestic reliance on coal—limited the Greenium (Panizza et al., 2025).

In Indonesia, a key event is the November 2022 G20 Summit in Bali, during which the Just Energy Transition Partnership (JETP) was announced, mobilizing approximately \$20 billion in climate financing. This event constitutes a Global Validation Shock, signaling strong international support for Indonesia’s energy transition. The empirical strategy tests whether this external endorsement generated pre-announcement drift in the Green Sukuk market, or whether concurrent domestic developments—such as the expansion of captive coal plants—introduced signaling noise that offset the positive effect and produced a Trust Penalty (Neumann, 2025).

In Thailand, the analysis focuses on the June 2024 Phase 2 Green Taxonomy update. This regulatory reform aimed to align Thailand’s classification framework with the EU Taxonomy, thereby enhancing transparency and reducing information asymmetry. Within the Credibility Gap framework, such a policy is expected to strengthen investor confidence. If the empirical results show a tightening of the Greenium (i.e., a more negative spread) following the update, this would provide evidence that consistent and transparent regulatory action enables governments to secure a sustained borrowing advantage.

These events are selected because they represent moments of intensified political signaling. By analyzing changes in the Greenium around these events, the study evaluates whether markets reward credible policy commitments with a lower cost of debt or penalize inconsistency through a Credibility Gap.

5 Data Analysis and Empirical Results

This section presents the empirical findings of the study, combining descriptive statistics, sentiment analysis, regression outputs, and validation tests to evaluate the relationship between political climate sentiment and sovereign Greenium dynamics.

5.1 Descriptive Statistics

This subsection presents a table reporting the average Greenium for India, Indonesia, and Thailand in both the pre-shock and post-shock periods. The objective is to provide a baseline comparison of borrowing cost dynamics across countries and identify initial patterns associated with political events.

5.2 Sentiment Analysis Results

This subsection visualizes the ClimateBERT sentiment scores through a high-frequency *Sentiment Timeline*. The analysis overlays key political speeches and policy announcements with movements in bond yields, allowing for a visual assessment of the relationship between political communication and financial market responses.

5.3 Regression Outputs

This subsection presents the formal results from the staggered Difference-in-Differences (DiD) regression model. The focus is on testing whether the key coefficients— β_1 (political sentiment) and β_2 (signaling noise)—are statistically significant.

The results are used to evaluate whether political communication has a measurable impact on the Greenium, and whether markets reward consistency (Credibility Premium) or penalize contradictory signals (Credibility Gap).

5.4 Parallel Trends Test

This subsection provides a graphical validation of the identification strategy. A key figure illustrates whether the green bond and its conventional twin exhibit similar trends prior to the political shock.

The objective is to demonstrate that the observed post-event changes are not driven by pre-existing differences, thereby supporting the causal interpretation of the results.

6 Discussion and IPE Interpretation

This section interprets the empirical findings through the theoretical framework developed in the literature review. It explains why the observed patterns in the Greenium occur, linking the results to concepts such as the Credibility Premium, Credibility Gap, and Signaling Noise.

6.1 Bridging the Credibility Gap

This subsection discusses cross-country differences in Greenium stability. It examines why certain countries, such as Thailand—with clearer regulatory frameworks—may exhibit a more stable Greenium, while others, such as Indonesia, experience volatility due to conflicting domestic political and industrial pressures.

6.2 The Cost of “Cheap Talk”

This subsection quantifies the financial cost of signaling noise. It translates changes in the Greenium into basis points and monetary terms, illustrating how inconsistent political communication

increases sovereign borrowing costs.

For example, a 4 basis-point increase on a \$1 billion bond implies an additional annual cost of approximately \$400,000, demonstrating that political credibility has direct fiscal consequences.

6.3 Global vs. Local Pressure

This subsection analyzes whether the Greenium responds more strongly to domestic political communication or to international climate events such as COP summits. The goal is to assess the relative importance of local versus global signaling in shaping investor perceptions.

7 Conclusion and Policy Recommendations

This section summarizes the main findings of the study and provides practical policy guidance for finance ministries in India, Indonesia, and Thailand.

7.1 Summary of Findings

This subsection provides a concise synthesis of the empirical results, emphasizing the role of high-frequency political sentiment in influencing sovereign borrowing costs and Greenium dynamics.

7.2 Policy Recommendations

This subsection offers practical recommendations for policymakers, focusing on the development of coordinated communication frameworks. In particular, it highlights the importance of alignment between institutions such as the Ministry of Finance and the Ministry of Energy to avoid contradictory policy signals and reduce signaling noise.

7.3 Limitations and Future Research

This subsection acknowledges the limitations of the study, including data availability constraints and potential measurement issues in sentiment analysis. It also outlines directions for future research, such as the use of more advanced artificial intelligence models and expanded datasets to improve the measurement of political credibility in financial markets.

References

Asian Development Bank. (2025). *AsianBondsOnline: Sovereign green bond database*. <https://asianbondsonline.adb.org/>

Baker, M., Bergstresser, D., Sergi, B., & Wurgler, J. (2022). Financing the response to climate change: The pricing and ownership of U.S. green bonds. *Critical Finance Review*, 11(3–4), 415–437.

Dragotto, M., Aristei, D., & Gallo, G. (2025). Climate awareness and green bond market dynamics: A high-frequency analysis. *International Review of Financial Analysis*, 91, 103012.

Flammer, C. (2021). Corporate green bonds. *Journal of Financial Economics*, 142(2), 499–516.

Gabor, D. (2021). The Wall Street Consensus. *Development and Change*, 52(3), 429–459.

International Capital Market Association. (2025). *Green bond principles: High-frequency disclosure guidelines*. ICMA Group.

Kling, G., Volz, U., Murinde, V., & Ayas, S. (2021). The impact of climate vulnerability on budgets and the cost of capital. *Ecological Economics*, 185, 107047.

Larcker, D. F., & Watts, E. M. (2020). Where is the greenium? *Journal of Accounting and Economics*, 69(2–3), 101312.

Lau, P., Sze, A., & Zhang, Y. (2024). Climate policy credibility and the pricing of green bonds. *Working paper*, Chinese University of Hong Kong.

Leetaru, K., & Schrod, P. (2013). GDELT: Global data on events, location, and tone, 1979–2012. *ISA Annual Convention*, 2, 1–49.

Neumann, T. (2025). *The political economy of green bonds in emerging markets: Coal interests and climate finance*. Springer Nature.

Panizza, U., Weder di Mauro, B., Shi, S., & Gulati, M. (2025). *The sovereign greenium: Big promise but small price effect* (IHEID Working Paper No. HEIDWP16-2025). Graduate Institute of International and Development Studies. https://repec.graduateinstitute.ch/pdfs/Working_papers/HEIDWP16-2025.pdf

Webersinke, N., Schimanski, T., Bingler, J., Leippold, M., & Kraus, M. (2022). ClimateBERT: A pre-trained language model for climate-related disclosures. *arXiv preprint arXiv:2110.12010*.

Zerbib, O. D. (2019). The effect of pro-environmental preferences on bond prices: Evidence from green bonds. *Journal of Banking & Finance*, 98, 39–60.